Project – Report

On Telecom DevOps Messaging Platform

Course Name: DevOps & Cloud Engineering

Institution Name: Medi-Caps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

In modern telecom systems, deploying applications manually is time-consuming, error-prone, and difficult to scale. There is a need for an automated system that ensures smooth deployment, faster delivery, and reliable infrastructure management.

1.2 Project Objectives

- To automate the software deployment lifecycle using CI/CD pipelines
- To containerize applications using Docker
- To deploy applications efficiently using Kubernetes
- To manage infrastructure using Terraform (Infrastructure as Code)
- To ensure scalability and reliability of telecom services

1.3 Scope of the Project

```
Plan → Code → Build → Test → Release → Deploy → Monitor
  ↑                                              ↓
  └──────────── Continuous Feedback ──────────────┘
```

- Covers end-to-end DevOps pipeline from code commit to deployment
- Focuses on containerized telecom messaging API
- Uses cloud-native tools for automation and orchestration
- Does not include advanced monitoring or security layers

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2. Proposed Solution

2.1 Key Features

- Automated CI/CD pipeline using Jenkins
- Docker-based containerization
- Kubernetes-based deployment and orchestration
- Terraform for infrastructure provisioning
- REST API for telecom messaging service

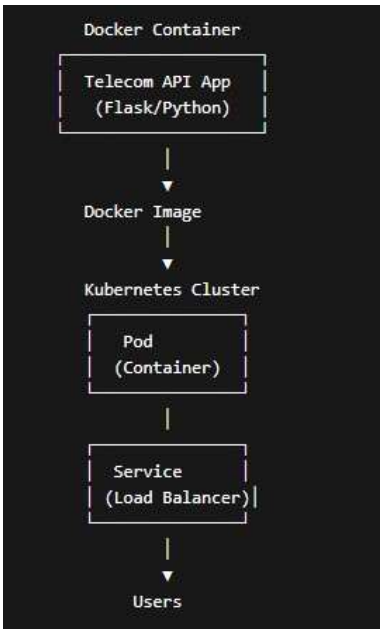
2.2 Overall Architecture / Workflow

```
Developer
   |
   ▼
GitHub (Code Push)
   |
   ▼
Jenkins (CI/CD Pipeline)
   |
   ├── Build Docker Image
   |
   ├── Run Tests
   |
   └── Push to Docker Hub
            |
            ▼
     Kubernetes Cluster
            |
            ▼
     Deploy Application
            |
            ▼
         Users / API Calls
```

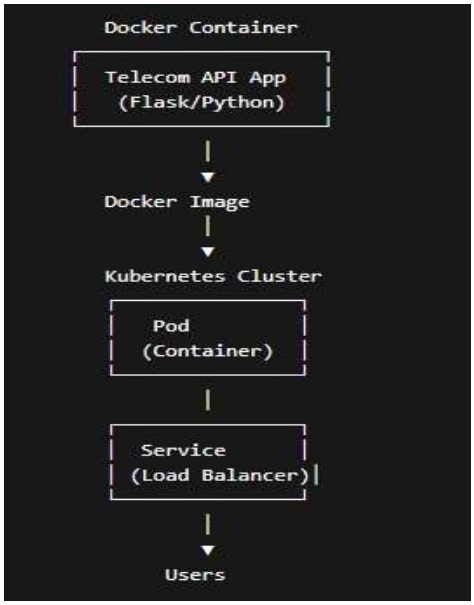
Workflow:

1. Developer pushes code to GitHub
2. Jenkins pipeline is triggered automatically
3. Docker image is built
4. Image is pushed to Docker Hub
5. Terraform provisions infrastructure
6. Kubernetes deploys the application

2.3 Tools & Technologies Used

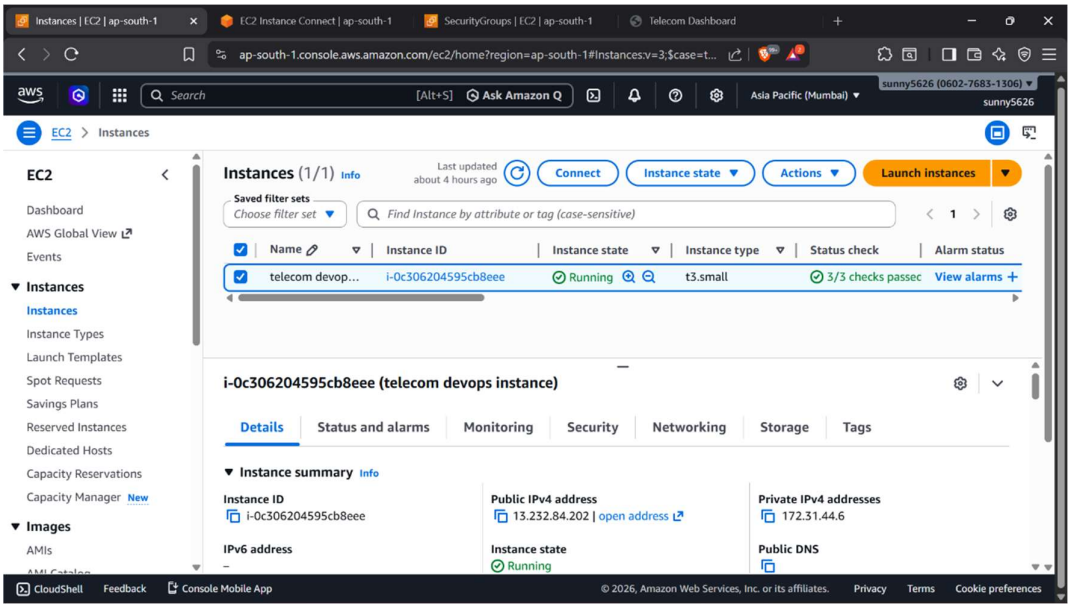


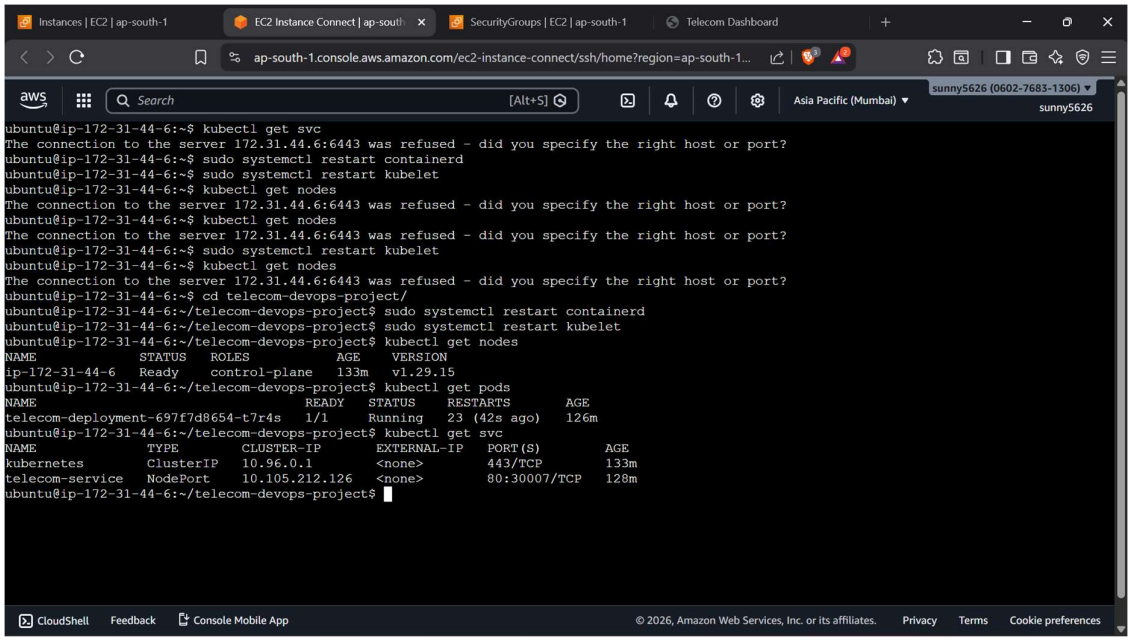
- Python (Flask)
- GitHub
- Jenkins
- Docker
- Docker Hub
- Kubernetes
- Terraform



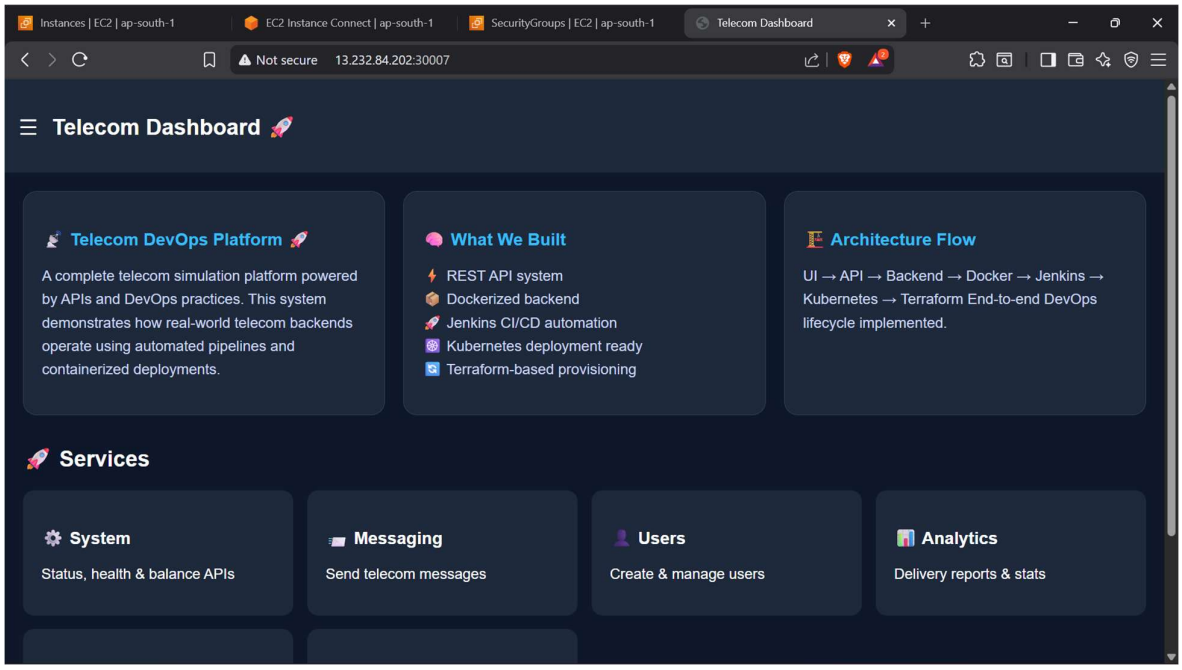
3. Results & Output

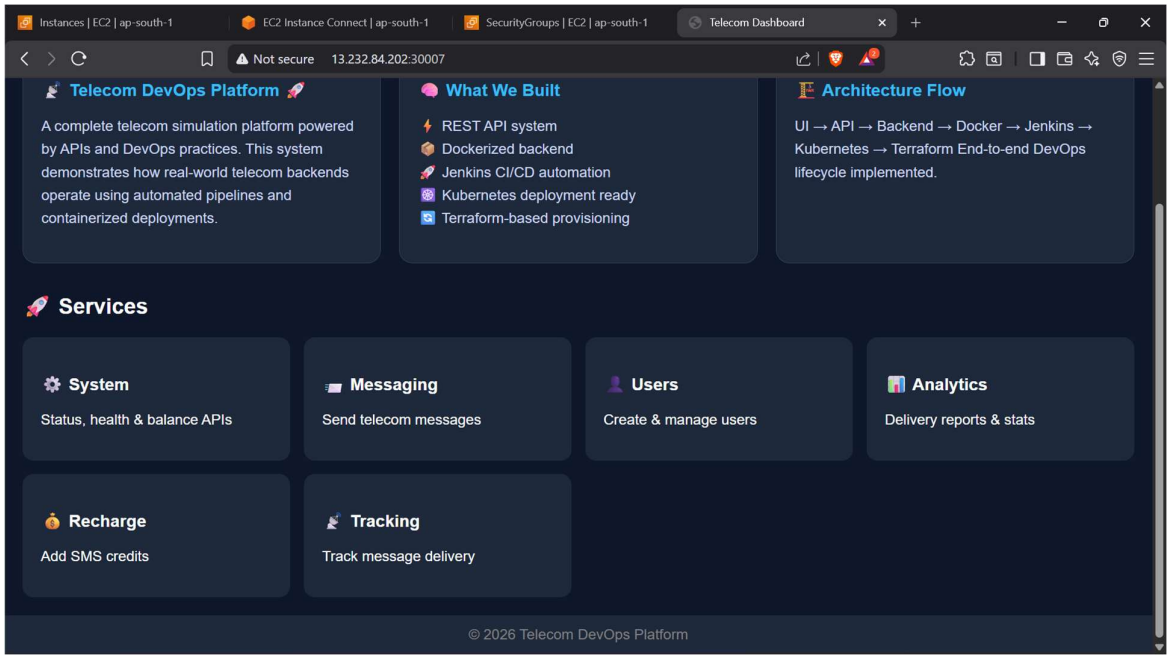
3.1 Screenshots / Outputs





3.2 Reports / Dashboards / Models





3.3 Key Outcomes

- Successfully automated deployment pipeline
- Reduced manual intervention
- Achieved scalable container deployment
- Improved system reliability and consistency

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4. Conclusion

This project successfully demonstrates a complete DevOps lifecycle for a telecom messaging API. By integrating tools like Docker, Kubernetes, Jenkins, and Terraform, we automated the deployment process from development to production. The project enhanced understanding of CI/CD pipelines, containerization, and infrastructure automation, making the system more efficient, scalable, and reliable.

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5. Future Scope & Enhancements

- Implement Kubernetes autoscaling
- Add monitoring tools like Prometheus and Grafana
- Use Helm for better Kubernetes management
- Integrate automated testing in CI/CD pipeline
- Improve security with DevSecOps practices